Week 5 BA

2024 Aug04 Quiz-2

Comprehension for questions 1-2 (source Q11)

A demand response curve has a constant elasticity of +0.4. If the price of the product is 100 and the corresponding demand is 200, then answer the given sub questions.

1. [Source Q12; SA]
What is the intercept for the curve? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 1261 to 1262]

2. [Source Q13; SA]
What should be the price to achieve a demand of 300? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 35 to 38]

Comprehension for questions 3-4 (source Q14)

The demand of a product is estimated to follow the relationship: D(p) = 1444 - 3(p^2). Here, "p" is the selling price for the product. The product is made at Rs. 50 per unit. Given this information, answer the given sub questions.

3. [Source Q15; SA]
If you intend to maximize the revenue, then what is optimal value for "p"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 12.50 to 12.70]

4. [Source Q16; SA]
If you intend to maximize the profits, then what is optimal value for "p"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 37 to 38]

Comprehension for questions 5-9 (source Q17)

"Fake Money (FM)" is an investment consulting firm. Dr. Milo has approached FM with Rs. 1,00,000 for guidance on investment opportunities. Dr. Milo wants to invest the entire Rs. 1,00,000. FM's top financial analyst recommends that all new investments be made in oil industry, steel industry or government bonds. Specifically, the analyst identified five investment opportunities and projected their annual rates of returns (which are specified in Table-1). Dr. Milo, is a guy who likes to distribute his investments. Hence, he has specified the following requirements to the analyst 1. Neither industry (oil or steel) should receive more than Rs. 50,000 2. Government bonds should be at least 25% of the steel industry investments 3. The investments in Bongu Oil, the high-return but high-risk investment, cannot be more than 60% of total oil industry investment Given this information, answer the given sub questions

Investment Opportunity	*Projected Annual Rate of Return (in %)*
Dhokha Oil	7.3
Bongu Oil	10.3
Achchha Steel	6.4
Nalla Steel	7.5
Government Bonds	4.5

5. [Source Q18; SA]
How many decision variable(s) is/are present in the standard form of the primal?

[Source ans: 1 or 5]

6. [Source Q19; SA]
How many constraints (excluding the non-negativity constraints) is/are present in the standard form of the primal?

[Source ans: 6]

7. [Source Q20; SA]
How many constraints (excluding the non-negativity constraints) are present in the dual (which is based on the standard form of the primal)?

[Source ans: 5]

8. [Source Q21; SA]

If FM has suggested the below distribution of investments, then how much money (total money) is projected to be in hand at the end of the first year? (Note: Enter the answer in "Rs." rounded to two decimal places without the "Rs." symbol. For example, if the answer is "Rs. 1.234", then enter it as "1.23")

Investment Opportunity	*Amount Invested (in Rs.)*
Dhokha Oil	20,000
Bongu Oil	30,000
Achchha Steel	0
Nalla Steel	40,000
Government Bonds	10,000

[Source ans: 108000]

9. [Source Q22; SA]

If FM has suggested the below distribution of investments, then how many decision variables in the dual will have a non-zero value?

Investment Opportunity	*Amount Invested (in Rs.)*
Dhokha Oil	20,000
Bongu Oil	30,000
Achchha Steel	0
Nalla Steel	40,000
Government Bonds	10,000

[Source ans: 5]

2024 Sep01 ET

Comprehension for questions 10-15 (source Q22)

Dr. Milo wants to advertise his new brand of tuition centres called "Torture Students". Currently promotion plans are underway for this year's admission. Advertising alternatives include TV, Radio and Social Media. The estimated audiences per advertisement (who will enrol after viewing the advertisement) and costs per advertisement for the different alternatives are shown in Table-2. To ensure a balanced use of advertisement media (1) radio advertisements must not exceed 70% of the total number of advertisements authorised (2) amount authorised for TV advertisements should account for at least 30% of the total spending (3) Amount of spending on social media advertisements cannot exceed more than 25% of the other two spending. The promotional budget is limited to Rs. 1,82,000, and Dr. Milo wants to use all the available money to maximize the number of enrolments. Then answer the given subquestions.

Component	*TV*	*Radio*	*Social Media*
Estimated audiences per advertisement (who will enrol after viewing the advertisement) (Numbers/ Advertisement)	4,000	1,800	10,000
Costs per advertisement (Rs./ Advertisement)	Rs. 2500	Rs. 400	Rs. 750

Table-2

10. [Source Q23; SA]
How many decision variables (count) are present in the standard form of the primal problem? (Note: Specify the count of variables. Not the count of notations)

[Source ans: 3]

11. [Source Q24; SA]
How many constraints (excluding the non-negativity constraints) are present in the standard form of the primal? (Note: Specify the count of constraints. Not the count of constrains in notation format)

[Source ans: 5]

12. [Source Q25; SA]
If the standard form of the primal is converted to a dual, then how many constraints (excluding the non-negativity constraints) are present in the dual? (Note: Specify the count of constraints. Not the count of constrains in notation format)

[Source ans: 3]

13. [Source Q26; MCQ]
If Dr. Milo has decided to invest Rs. 1,05,000 in TV and Rs. 31,500 in Social Media and Rs. 45,200 in Radio. Then has he acted like a true operations expert and found a Feasible solution?

(a) Yes, he is a great operations expert as the above is a feasible solution

(b) No, he is not an expert but a fraud as the above is NOT a feasible solution

[Source ans: b]

14. [Source Q27; SA]
For the sake of argument, assume that Dr. Milo is an operations expert (irrespective of your answer to the previous question). Then, for the solution (Rs. 1,05,000 in TV and Rs. 31,500 in Social Media and Rs. 45,200 in Radio are made) how many (count) decision variables in the dual (which is formulated based on the standard form of the primal) will have a non-zero value?

[Source ans: 0]

15. [Source Q28; SA]
For the sake of argument, assume that Dr. Milo is an operations expert (irrespective of your answer to the previous question). Then, for the solution (where an investment of Rs. 1,05,000 in TV and Rs. 31,500 in Social Media and Rs. 45,200 in Radio are made) what is the objective function value for the standard form of the primal?

[Source ans: -791400]

2024 Dec01 Quiz-2

16. [Source Q4; SA]
The relationship between Demand "D" and Selling Price "P" is given by the equation D(p) = 720 - 5*P. If the intention is to maximize the profit, then what is the optimal selling price if the item is going to be made at Rs. 20 per unit?

[Source ans: 82]

17. [Source Q5; SA]
Say a demand response curve is modelled as a constant elasticity curve. If Q1 is 2400 units, Q2 is 1500 units, P1 is Rs. 100 and P2 is Rs. 200, then what is the elasticity of the curve? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 0.65 to 0.70]

Comprehension for questions 18-21 (source Q22)

The following is the Primal LP Formulation for a resource allocation problem. Given this, answer the subquestions based on the standard form of the primal

Objective: Maximize $(10Y_1 + 20Y_2 + 5Y_3 + 7Y_4 + 12Y_5 + 11Y_6)$

Constraint-1: $Y_1+Y_2+Y_3+Y_4+Y_5+Y_6 <= 200$

Constraint-2: $2Y_1+3Y_3 <= 50$

Constraint-3: $4Y_2+5Y_4+3Y_6 = 65$

Constraint-4: $Y_1+ Y_3 <= 20$

Constraint-5: $Y_2+Y_4+Y_6 = 16$

Constraint-6: $Y_1, Y_2, Y_3, Y_4, Y_5, Y_6$ are all non-negative

18. [Source Q23; SA]
How many constraints will be present in the dual formulation?

[Source ans: 6]

19. [Source Q24; SA]
How many decision variables will be required to formulate the dual?

[Source ans: 7]

20. [Source Q25; SA]
What is the objective function "co-efficient" for the dual decision variable corresponding to the "Constraint-4" in the primal?

[Source ans: 20]

21. [Source Q26; SA]
For the following solution of the primal, how many decision variables in the dual will have a NON- ZERO value ? Y1= 4 Y2= 5 Y3= 2 Y4 = 6 Y5= 10 Y6= 5

[Source ans: 4]

2024 Dec22 ET

Comprehension for questions 22-25 (source Q3)

Milo's Survey Services (MSS) specializes in evaluating the reaction of customers to new products. A client has hired MSS to ascertain customer reaction to a recently launched product (Frying Pan). During meeting with the client, MSS agreed to conduct door-to-door personal interviews to obtain responses from households with children and without children. In addition, MSS agreed to conduct both "Daytime" and "Evening time" interviews. Specifically, the client's contract called for MSS to conduct 1000 interviews under the following quota guidelines i. Interview at least 400 households with children ii. Interview at least 400 households without children iii. The total number of households interviewed during evening must be at least as great as the number of households interviewed during the day iv. At least 40% of the interviews for households with children must be conducted during the evening v. At least 60% of the interviews for households without children must be conducted during evening Because the interviews for households with children take additional interview time and because evening interviewers are paid more than daytime, the cost varies with type of interview. Based on previous surveys conducted, MSS estimates the interview costs as given in Table-1 MSS wanted to conduct the interviews so that the overall cost for conducting the survey is minimized. Given this information, answer the subquestions.

Type of Household	Interview Cost	
	Daytime	Evening Time
With Children	₹20	₹25
Without Children	₹18	₹20

Table-1

22. [Source Q4; SA]
How many (count of) decision variables are present in the standard form of the primal?

[Source ans: 4]

23. [Source Q5; SA]
Excluding the non-negativity constraint, how many (count of) constraints are present in the standard form of the primal?

[Source ans: 7]

24. [Source Q6; SA]
MSS has decided to conduct 400 interviews for Households with children and 600 interviews for households without children. Moreover, of the 400, 240 are to be conducted in the daytime. Similarly, of the 600 interviews, 360 are going to be conducted in the evening. Then, what is the total cost that would be incurred by MSS? (Note: enter your answer in "Rupees" rounded to two decimal places without the "Rs." symbol. For example, if your answer is "Rs.123.456" then enter it as "123.46")

[Source ans: 20320 or -20320]

25. [Source Q7; SA]
MSS has decided to conduct 400 interviews for Households with children and 600 interviews for households without children. Moreover, of the 400, 240 are to be conducted in the daytime. Similarly, of the 600 interviews, 360 are going to be conducted in the evening. Then, how many (count of) decision variables will have a NON-ZERO VALUE in dual formulation which is obtained based on the standard form of the primal?

[Source ans: 2]

2025 Mar16 Quiz-2

26. [Source Q3; SA]

Source limitation: the nonlinear curve family is unspecified; the model cannot be uniquely determined.

The demand-response curve for a product is expected to be non-linear curve. It is noted that the demand was 1000 units at a price of Rs.80 and 2500 units at a price of Rs. 40. Then, at a price of Rs. 60, what is the expected demand? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "10.235", then enter it as "10.24")

[Source ans: 1455 to 1470]

27. [Source Q5; MSQ]

You solve the primal of a linear program with a maximization objective, three decision variables and two constraints of the less than or equal to type. Non-negativity restrictions apply to the decision variables. After solving the linear program, you find that the first constraint is not binding (LHS $<$ RHS) and the second constraint is binding (LHS $=$ RHS). Which of the following statements is/are correct?

(a) There are three decision variables in the dual

(b) The dual variable corresponding to the second constraint is zero

(c) There are two decision variables in the dual formulation

(d) The dual variable corresponding to the second constraint is non-zero

[Source ans: c, d]

Comprehension for questions 28-31 (source Q6)

(The following comprehension is purely imaginary) "FUNKY" is a compay that manufactures purses, shoes and backpacks. The construction of the three products requires leather and synthetic material, with leather being the limiting raw material. The production process uses two types of skilled labour: Sewing and Polishing. Table-2 gives the availability of the resources, the usage in the three products and the profits per unit. Given the above information, formulate the problem as a linear program to maximize the profit and answer the subquestions.

Resource	Resource requirements per unit			Daily availability of resource
	Purse	Shoes	Backpacks	
Leather (in feet)	2	1	3	42
Sewing (in Hours)	2	1	2	40
Polishing (in Hours)	1	2	1	42
Profit (in Rs./unit)	24	22	45	--

Table-2

28. [Source Q7; SA]

How many (count of) decision variables are present in the standard form of the primal?

[Source ans: 3]

29. [Source Q8; SA]

Excluding the non-negativity constraint, how many (count of) constraints are present in the standard form of the primal?

[Source ans: 3]

30. [Source Q9; SA]
If you decide to make and sell 10 Purses, 15 Shoes and 2 Backpacks, then how many (count of) decision variables in the dual (which is developed based on the standard form of the primal) will have a value of "0"?

[Source ans: 2]

31. [Source Q10; SA]
If you decide to make and sell 10 Purses, 15 Shoes and 2 Backpacks, then what is the primal objective function value?

[Source ans: 660]

Comprehension for questions 32-33 (source Q11)

The estimated demand for a product follows the relationship: D(p) = 2570 - 4p. Where "p" is the selling price of the product. With this information answer the given subquestions.

32. [Source Q12; SA]
If you intend to maximize the revenue, what is the best value for "p"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "10.235", then enter it as "10.24")

[Source ans: 321 to 322]

33. [Source Q13; SA]
It costs Rs. 250 to make the product. Then, if you intend to maximize the profit, what is the best value for "p"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "10.235", then enter it as "10.24")

[Source ans: 446 to 447]

2025 Apr13 ET

Comprehension for questions 34-35 (source Q27)

(The following is a purely imaginary scenario) The relationship between Demand "D" and Selling Price "P" is given by the equation D(p) = 780 - 9*P. Then answer the given sub-questions.

34. [Source Q28; SA]
If the intention is to maximize the profit, then what is the optimal selling price if the item is going to be made at Rs. 30 per unit? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 58.00 to 59.00]

35. [Source Q29; SA]
What is the maximum profit that can be generated? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 20424.00 to 20425.00]

2025 Aug03 Quiz-2

Comprehension for question 36 (source Q2)

Data on "Urban Population (UP)" of a country and the "GDP" of the same country are collected. The "GDP" is captured as a percentage (that is "10.5%" is captured as "10.5") and the "UP" is captured in crores of people (that is "1,20,00,000" is captured as "1.2"). From the raw data, the analyst has identified that the relation is non-linear. Hence, the analyst applies a natural logarithmic transformation on both variables (the independent variable is "GDP" and the dependent variable is "UP"). The transformed data is modelled as a Simple Linear Regression and the developed model is presented in Figure-1. Using this information, answer the given subquestions.

	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
Ln(GPD)	10.43004	.8278521	12.599	0.000	8.792235	12.06785
_cons	-24.42095	6.295892	-3.879	0.000	-36.87662	-11.96528

Figure-1

36. [Source Q4; SA]
If the GDP in 2024 was 15%, then what is the predicted "UP" (in crores of people) based on the SLR model? (Note: Enter your answer rounded to two decimal places. For example, if your answer is "1.235" then enter it as "1.24")

[Source ans: 44 to 46]

Comprehension for questions 37-38 (source Q21)

A demand response curve has a constant price elasticity of +0.6. If the price of the product is 100 and the corresponding demand is 200, then answer the given subquestions.

37. [Source Q22; SA]
What is the value of "Constant" ("C") for the constant elasticity model? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 3169 to 3170]

38. [Source Q23; SA]
What should be the price to achieve a demand of 300? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 50 to 60]

2025 Aug31 ET

Comprehension for questions 39-43 (source Q17)

The Following Comprehension and Related Subquestions are Purely Hypothetical Milo The Goofball (MTG) is a company that makes footballs for "Recreation (R)" games and "Professional (P)" games. The process of making a football involves the following three stages Stage-1: Cutting and Dyeing Stage-2: Sewing Stage-3: Inspection It is seen that making a R-game football requires 6 minutes in Stage-1, 15 minutes in Stage-2 and 6 minutes in Stage-3. On the other hand, the football for a P-game requires 12 minutes, 9 minutes and 6 minutes in Stages 1,2 and 3 respectively. For the coming planning period of 4 weeks, MTG has 340 hours available in Stage-1, 480 hours available in Stage-2 and 300 hours available in Stage-3. Moreover, if required, MTG can avail more hours in Stage-1 at an additional cost of Rs. 200 per hour. Moreover, a R-game football generates a profit of Rs. 500 per piece and a P-game football generates a profit of Rs. 800 per piece. Furthermore, MTG has forecasted that the maximum number of footballs it can sell is 3000. With an objective to maximize the profit, formulate the problem as a Linear Program and answer the given subquestions.

39. [Source Q18; SA]
How many decision variables are present in the standard form of the primal linear Programming Problem?

[Source ans: 3]

40. [Source Q19; SA]
How many decision variables are present in the Dual Linear Programming Problem (which is developed based on the standard form of the problem)?

[Source ans: 4]

41. [Source Q20; SA]
If the company decides to make 1500 P-game footballs, 1000 R-game footballs, then how many decision variables in the Dual Linear Programming Problem (which is developed based on the standard form of the problem) will have a value of "0"?

[Source ans: 1]

42. [Source Q21; SA]
If the company decides to make 1500 P-game footballs, 1000 R-game footballs, then what is the objective function value for the primal linear programming problem (which is in the standard form)? [Note: Enter your answer rounded to two decimal places. For example, if your answer is 1.235 then enter the answer as "1.24"]

[Source ans: -1238000.00 or 1238000.00]

43. [Source Q22; MSQ]
If the MTG decides to make 1500 P-game footballs, 1000 R-game footballs, then which of the following options is/are correct

(a) MTG will purchase addition capacity in Stage-1

(b) MTG will not purchase addition capacity in Stage-1

(c) MTG will have excess capacity in Stage-1

(d) MTG will have excess capacity in Stage-2

(e) MTG will have excess capacity in Stage-3

(f) MTG will have deficit capacity in Stage-2

(g) MTG will have deficit capacity in Stage-3

(h) MTG cannot make 1500 P-game footballs and 100 R-game footballs as it is aninfeasible solution

(i) Cannot say as more information is required

[Source ans: a, d, e]

2026 Apr06 Quiz-2

Comprehension for questions 44-45 (source Q10)

ABC is a company that produces three products: "Phone", "TV", and "Computer". The production of each product involves four operations: "Welding", "Cutting", "Soldering" and "Inspection". Table-3 provides information relating to the manufacturing requirements. The total working hours available for production in a day is 10 hours, and all units produced can be sold. Given this information, answer the given subquestions

Product	*Minutes Required in the Operation*				*Profit made in selling the product (in INR)*
	Welding	*Cutting*	*Soldering*	*Inspection*	
Phone	3	2	4	5	300
TV	20	24	22	7	700
Computer	30	17	15	9	500

Table-3

44. [Source Q11; SA]
If the objective is to maximize profit, then how many decision variables would be present based on the standard (canonical) primal formulation?

[Source ans: 3]

45. [Source Q12; SA]
Based on the standard (canonical) primal formulation, in the dual formulation how many decision variables will be present?

[Source ans: 1]

2026 May06 ET

Comprehension for questions 46-51 (source Q34)

The following scenario is purely hypothetical Dr. Milo wants to advertise his new brand of turion centres called "Torture Students". Currently promotion plans are underway for this year's admission. Advertising alternatives include TV, Radio and Social Media. The estimated audiences per advertisement (who will enrol after viewing the advertisement) and costs per advertisement for the different alternatives are shown in Table-3. To ensure a balanced use of advertisement media (1) radio advertisements must not exceed 70% of the total number of advertisements authorised (2) amount authorised for TV advertisements should account for at least 30% of the total spending (3) Amount of spending on social media advertisements cannot exceed more than 25% of the other two spending. The promotional budget is limited to Rs. 1,82,000, and Dr. Milo wants to use all the available money to maximize the number of enrolments. Then answer the given subquestions.

Component	*TV*	*Radio*	*Social Media*
Estimated audiences per advertisement (who will enrol after viewing the advertisement) (Numbers/ Advertisement)	4,000	1,800	10,000
Costs per advertisement (Rs./ Advertisement)	Rs. 2500	Rs. 400	Rs. 750

Table-3

46. [Source Q35; SA]
How many decision variables (count) are present in the standard (canonical) form of the primal problem? (Note: Specify the count of variables. Not the count of notations)

[Source ans: 3]

47. [Source Q36; SA]
How many constraints (excluding the non-negativity constraints) are present in the standard (canonical) form of the primal? (Note: Specify the count of constraints. Not the count of constrains in notation format)

[Source ans: 5]

48. [Source Q37; SA]
If the standard (canonical) form of the primal is converted to a dual, then how many constraints (excluding the non-negativity constraints) are present in the dual? (Note: Specify the count of constraints. Not the count of constrains in notation format)

[Source ans: 3]

49. [Source Q38; MCQ]
If Dr. Milo has decided to invest Rs. 1,05,000 in TV and Rs. 32,000 in Social Media and Rs. 45,000 in Radio. Then has he acted like a true operations expect and found a Feasible solution?

(a) Yes, he is a great operations expert as the above is a feasible solution

(b) No, he is not an expert but a fraud as the above is NOT a feasible solution

[Source ans: a]

50. [Source Q39; SA]
For the sake of argument, assume that Dr. Milo is an operations expert (irrespective of your answer to any of the previous questions). Then, for the solution (Rs. 1,05,000 in TV and Rs. 32,000 in Social Media and Rs. 45,000 in Radio are made) how many (count) decision variables in the dual (which is formulated based on the standard (canonical) form of the primal) will have a non-zero value?

[Source ans: 2]

51. [Source Q40; SA]
For the sake of argument, assume that Dr. Milo is an operations expert (irrespective of your answer to any of the previous questions). Then, for the solution (Rs. 1,05,000 in TV and Rs. 32,000 in Social Media and Rs. 45,000 in Radio are made) what is the objective function value the primal formulation?

[Source ans: 912000]